

Spectrum regulation: Crucial to private wireless deployment

The allocation and use of private wireless networks are intrinsically linked with spectrum bands' availability and regulatory permissions. The following points highlight key aspects of this issue:

- Private LTE networks have successfully utilised unlicensed spectrum, such as the 5.8 gigahertz band used in the industrial proof-of-concepts.
- The transition towards 5G has prompted discussions with regulatory bodies like the Department of Telecommunications for the allocation of unlicensed spectrum.
- These negotiations aim to secure permissions similar to those granted for ham radios to facilitate private cellular implementations.
- While discussions for 5G are still ongoing, private LTE has already seen successful demonstrations, such as in Nokia's train-based private LTE trials.

In summary, the regulation and allocation of spectrum bands play a critical role in the deployment and success of private wireless networks. While LTE has made headway in this area, the conversation around 5G's allocation is still unfolding.

the aluminium and copper industry valued at billions of US dollars, faced significant operational challenges. Situated in an advanced facility with some of Asia's most sophisticated equipment, their operational demands required low latency, expansive coverage, and high reliability in monitoring. Although sections of their facility had Wi-Fi, it suffered from bandwidth limitations, latency issues, and poor penetration. The challenge was clear: the need for a resilient network solution compatible with high-mobility equipment.

In such circumstances, a guided process was initiated to lead them through a transformative experience. The process commenced with a demonstration of the concept, focused on LTE technology, with a deliberate choice of private LTE over 5G. As the chosen technology partner, Nokia ensured the private wireless setup integrated seamlessly with various industrial protocols. Through Siemens programmable logic controller (PLC), a successful private cellular network was established, enabling real-time monitoring of pivotal parameters like temperature and speed for various equipment. "This implementation was executed utilising the unlicensed band of 5.8 gigahertz of

LTE, a spectrum free for public use," says Abin.

An extensive industrial complex in the US. A different challenge took centre stage at a large industrial complex in Texas, United States. Despite its size, encompassing over 46.6 square kilometres (18 square miles) with more than 35 production plants, the complex grappled with inefficiencies stemming from paper-based processes and security concerns. Maintenance tasks were tedious and manual, often leading to significant time losses.

To address these inefficiencies, a digital transformation was initiated. Solutions integrated technologies from industry stalwarts like Nokia, Microsoft, and Palo Alto Networks, providing the industrial complex with devices designed for their industrial context. Enhanced communication became a reality with the inclusion of platforms like Microsoft Teams. Augmented reality applications under development would offer a digital twin representation of the physical infrastructure, effectively minimising maintenance downtime.

The expansive coverage requirements were met through a private LTE network, enveloping the entire Texas complex within five months. Utilising the CBRS (a shared spec-

trum available in the US), a wireless device control system was deployed to manage device registration and ensure secure access. Push-to-talk capabilities were also introduced, facilitating uninterrupted communication for workers even in challenging environments. The outcome was remarkable. "In just four months, around 28,000 manual processes underwent digitisation here, leading to streamlined operations, enhanced efficiency, and reduced manual errors," says Abin.

In a rapidly evolving technological landscape, the advent and rise of private wireless networks underscore the inevitable shift towards digitised industrial processes. While traditional wireless solutions like Wi-Fi continue to have their place, the unique challenges of today's industries—ranging from the need for ultra-low latency to massive device connectivity—highlight private wireless's value proposition. This is especially evident when considering mission-critical applications, expansive coverage requirements, and cost efficiencies. Recent advancements and practical implementations in diverse sectors further emphasise the transformative potential of private wireless solutions. With tech giants and industrial behemoths moving in unison, integrating private wireless technology is not just a fleeting trend but an intrinsic part of Industry 4.0. The real-world impact, demonstrated in both the metal production industry in India and the extensive industrial complex in Texas, is a testament to its potential to redefine operational processes. The future of industries is intrinsically tied to the evolution and adoption of private wireless networks. **EFY**

This article is compiled from a tech talk session at EFY EXPO 2023, Pune by Abin Chacko Moozhijil, Global Marketing Practice Leader, Network & Edge, Kyndryl. It has been transcribed and curated by Nidhi Agarwal, Technology Journalist at EFY.